SECTION–I(i) : (Maximum Marks : 24)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

1. Which of the following sodium salt react with dil. H_2SO_4 & gives triatomic gas

(A) Na_2CO_3 (B) Na_2SO_3 (C) Na_2S (D) All of the above

2. In which of the following option N_2 gas is formed

(A) $Mg_3N_2 + H_2O \longrightarrow$ (B) $NH_4NO_3 \xrightarrow{\Delta}$

(C) $Zn + dil.HNO_3 \longrightarrow$ (D) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$

3. Which of the following salts of anion produce coloured vapours with conc. H_2SO_4

(A) NaCl (B) Na_2S (C) $Na_2S_2O_3$ (D) NaI

4. $3BCl_3 + 3\underline{N}H_4Cl \xrightarrow{140°C} \underline{B}_3N_3H_3Cl_3 \xrightarrow{NaBH_4} \underline{B}_3N_3H_6 \xrightarrow{3HCl} \underline{B}_3N_3H_9Cl_3$.

The protonation in the last step occurs on :

(A) B-atom (B) N-atom
(C) Cl-atom (D) Can not be predicted

5. Which of the following cation does not form soluble complex with excess of NH_3?

(A) Fe^{+3} (B) Zn^{+2} (C) Cu^{+2} (D) Ag^{+}

6. Which of the following element does not show –3 oxidation state.

(A) Bi (B) N (C) As (D) None of these

7. When H_2O_2 reacts with HIO_4 & NH_2OH separately, what are the iodine containing product & nitrogen containing product respectively.

(A) I^-, NH_3 (B) IO_3^-,NO_3^- (C) IO_3^-,N_2O (D) I_2, N_2

8. Which of the following set is **CORRECT** matching.

	Compound		**Correct Match**
(A)	CO	$\longrightarrow$	Ethanol amine is correct absorber
(B)	H_2S	$\longrightarrow$	H_2SO_4 (conc.) is correct drying agent
(C)	NH_3	$\longrightarrow$	Anhydrous $CaCl_2$ is correct drying agent
(D)	PH_3	$\longrightarrow$	$HgCl_2$ solution is correct absorber

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SECTION-I(ii) : (Maximum Marks: 28)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

9. Which of the following oxy acid(s) on heating form one of the products in the lowest oxidation state of underlined atom when disproportionation reaction takes place.

(A) $H_3\underline{P}O_2$ (B) $H_3\underline{P}O_3$ (C) $H\underline{N}O_2$ (D) $H_2\underline{S}O_4$

10. In large scale NH_3 is prepared by Haber's process. Select the **CORRECT** statement(s) for Haber's process

(A) ΔH_f° for NH_3 is –ve

(B) iron oxide with small amount $K_2O + Al_2O_3$ is used as catalyst

(C) high pressure favour the formation of NH_3

(D) Extremly high temperature favour the formation of $NH_3 \approx 2700°C$

11. Which of the following statement(s) is/are **CORRECT**?

(A) In Ostwald process for the preparation of HNO_3, Pt/Rh is used at catalyst.

(B) In Contact process for the preparation of H_2SO_4, V_2O_5 is used as catalyst.

(C) In Decan process for the preparation of Cl_2, $CuCl_2$ is used as catalyst.

(D) In Haber process for the preparation of NH_3, (iron oxide, Al_2O_3, K_2O) is used as catalyst.

12. $NH_4ClO_4 \xrightarrow{\Delta} X\uparrow$, Gas (X) is produced by heating -

(A) $(NH_4)_2CO_3$ (B) $(NH_4)_2SO_4$ (C) $(NH_4)_2Cr_2O_7$ (D) NH_4NO_2

13. Consider the following statement(s) number of **CORRECT** statement is/are

(A) Anhydride of HNO_3 is an acidic oxide

(B) $H_2S_2O_3$, $H_2S_2O_4$, $H_2S_2O_5$, $H_2S_xO_6$ all contain atleast one S-S linkage

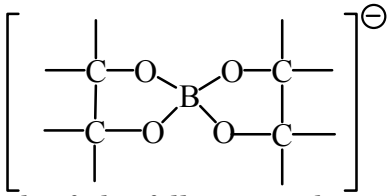
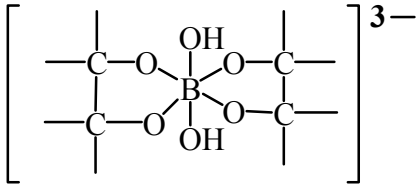
(C) Carbon monoxide is produced along with pure ZnO on heating basic Zinc Carbonate.

(D) All oxides of Sulphur and Chromium are acidic

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14. On addition of Cis 1-2 diol in the aq. solution of boric acid -

(A) H^+ ion concentration increases (B) H^+ ion concentration decreases

(C)  is formed (D)  is formed

15. Which of the following substance is/are identified due to their unique characteristic smell.

(A) PH_3 (B) O_3 (C) NH_3 (D) H_2S

SECTION–I(iii) : (Maximum Marks : 12)

- This section contains **ONE** paragraph.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.

Zero Marks : 0 In all other cases.

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(A) highly reactive white solid has garlic smell $\xrightarrow{\text{burns in excess air}}$ (B) strong dehydrating agent $\xrightarrow{\text{conc. } HNO_3}$ (F) + (C) colourless solid $\xrightarrow{I_2}$ (D) solid + E(g)

16. When (C) reacts with NaCl, the state of hybridisation of N-atom in the products.

(A) sp^2 only (B) sp^2 and sp^3 (C) sp^3 only (D) sp and sp^2

17. Which of the following statements is/are **INCORRECT**

(A) 'E' produces HNO_2 and HNO_3 when reacts with water.

(B) 'D' is used to estimate CO gas

(C) 'B' has cage like structure

(D) None of these

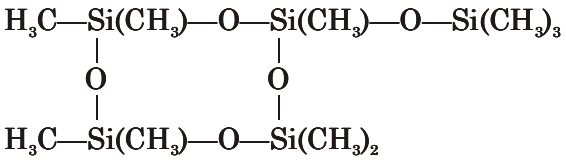
SECTION–IV : (Maximum Marks : 09)

- This section contains **THREE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.

Zero Marks : 0 In all other cases.

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1. 

Under hydrolytic condition, the number of the following compounds (R_nSiCl_{4-n}) which is/are used to prepare given silicon's compound(s).

$Si(CH_3)_4$, $(CH_3)_3SiCl$, $(CH_3)_2SiCl_2$, $(CH_3)SiCl_3$, $SiCl_4$

2. How many compounds liberate NH_3 on heating from the following.

$(NH_4)_2SO_4$ $(NH_4)_2CO_3$ NH_4Cl NH_4NO_3 NH_4NO_2

3. XeF_4 reacts with PF_5 as follows :

$XeF_4 + PF_5 \rightarrow [XeF_3]^+ [PF_6]^-$

If total numbers of 90° angles in reactants = x and if total number of 90° angles in products = y, considering all effects of VSEPR theory, then calculate the value of (y –x).

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